

The Basics of Graphic Network Simulator 3

GNS3 is a software emulator for networks. The GNS3 website describes it as the '...software that empowers network professionals.' This article provides the reader with the basics of GNS3 and a few test labs, which could be tried out by interested readers.



Reconfiguring a live network - for example, while implementing access control lists (ACLs) in an enterprise network - can be a challenging task. Troubleshooting such networks could also be really difficult if strict change management policies are in place, and specially if downtime and disruption are to be avoided under any circumstances.

Is there a way out of this? Yes! Many professionals use GNS3 to simulate complex networks. It is also used by students to prepare for lab tests conducted by various certification providers.

History

In 2005, Christophe Fillot wrote Dynamips - a Cisco router simulator. It had a text based interface and could simulate only one router on one PC, at a time. Version 0.2.5 of

Dynamips, released in 2006, introduced hypervisor mode and enabled multiple router simulations on one PC. It also provided a simple virtual switch.

GNS3 was introduced in September 2007 with a GUI, drag-and-drop facilities for adding network components, and easy storage of created objects and topology.

gns3.com describes GNS3 as 'The software that empowers network professionals.'

Dependencies

GNS3 depends on several libraries and components. Successfully installing these dependencies is a prerequisite. The important dependencies are listed below.

- Dynamips: Base of GNS3, a Cisco router simulator.
- Dynagen: A text based front-end to Dynamips, which uses the hypervisor mode to communicate with Dynamips.